

Epithelium Lab & Fundamentals of Histology and Pathology

Faculty pathology deck primary - official Week 4 SAL objectives integrated - microscopy, Proscia, epithelial/connective tissue recognition, tissue preparation, stains, and core histopathology language

9 official objectives

Microscopy

Proscia

Epithelium

Connective tissue

H&E + special stains

Dysplasia / invasion

SOURCE PRIORITY: the uploaded Barrett *Fundamentals of Pathology* deck controls lecture terminology, examples, and pathology emphasis. The official Week 4 learning-objective sheet defines the lab skills and tissue categories that must be mastered. Items such as microscope operation and the full connective-tissue list are therefore included as objective-driven lab skills even when the lecture deck itself is brief on them.

Official Week 4 SAL Objectives

1. Identify the parts of a light microscope and explain the operation of a microscope for viewing glass slides.
2. Demonstrate familiarity with Proscia software for viewing digital slides.
3. Identify simple squamous epithelium and distinguish the terms endothelium and mesothelium.
4. Identify simple cuboidal epithelium, locate its basement membrane, and explain basement-membrane function and appearance.
5. Identify simple columnar, pseudostratified columnar, and stratified squamous epithelia; identify cilia/goblet cells and explain how plane of section alters appearance.

6. Identify loose (areolar), dense irregular, dense regular, elastic, and adipose connective tissue and discuss composition/function.

7. Compare normal tissue with tissue involved by pathologic processes.

8. Recognize basic stains and preparation methods used to evaluate cellular macromolecules and tissue specimens.

9. Apply basic histopathology terminology and recognize techniques used for direct tissue sampling.

SAL Skill 1

Light microscope: orient first, then increase magnification

LO 1

Core parts

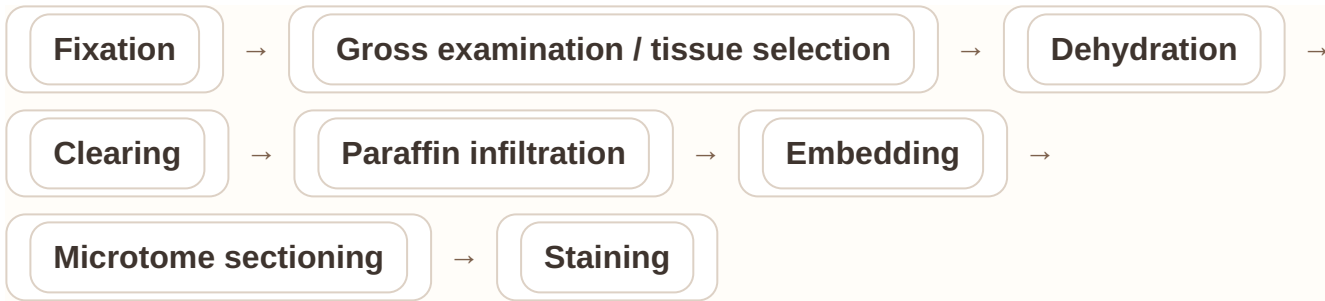
Eyepiece/ocular, objective lenses, revolving nosepiece, stage/mechanical stage, condenser, iris diaphragm, light source, coarse-focus control, and fine-focus control.

Safe workflow

Start on the lowest-power objective, center the specimen, bring the image into focus, then move to higher power. Use fine focus at high magnification; avoid driving the objective into the slide.

Field-of-view rule: increasing objective magnification decreases the field of view and depth of field. Always locate the structure at low power before zooming in.

Plane-of-section trap: a three-dimensional tube cut transversely can look circular, while the same tube cut obliquely/longitudinally looks elongated. Do not reclassify an epithelium solely from one oddly oriented profile.



Fixation

The deck emphasizes buffered formalin as the usual fixative. Fixation preserves tissue architecture and limits degradation.

Grossing and selection

Specimens are weighed/measured, serially sectioned, palpated, photographed when useful, and sampled from key areas. The goal is to document the lesion and its relationship to surrounding structures.

Embedding and sectioning

Processed tissue is embedded in paraffin wax, then a microtome cuts thin sections (deck: approximately 4-10 μm). Histology is a 2-D slice of a 3-D structure, so orientation matters.

White space is information: it may represent a true lumen/void or material lost during routine processing. Fat is commonly dissolved during clearing, so adipocytes often appear as large empty spaces with thin peripheral cytoplasm.

Method	What it highlights	High-yield clue
Hematoxylin	Acidic/negatively charged structures: DNA, RNA, ribosome-rich RER	Blue-purple / basophilic
Eosin	Basic/protein-rich structures: cytoplasm, collagen	Pink-red / eosinophilic
PAS	Glycogen, neutral mucins, basement membranes, fungal cell walls	Magenta
PAS + diastase	Diastase digests glycogen	Loss of PAS signal identifies glycogen component
Trichrome	Type I collagen	Useful for fibrosis / connective tissue
Reticulin	Type III collagen / reticulin fibers	Black network
Elastic stain	Elastic fibers / laminae	Highlights elastic architecture
Silver stains	Selected fungi, spirochetes, basement membranes	Organism/structure enhancement
Gram / acid-fast / Giemsa	Bacteria / mycobacteria / selected parasites & organisms	Microorganism-oriented stains
Prussian blue / von Kossa	Iron / calcium deposits	Pigment-mineral identification

IHC / IF

Antibodies detect specific antigens. Immunohistochemistry produces a visible chromogenic reaction; immunofluorescence uses a fluorescent label.

Electron microscopy

Electrons have a shorter wavelength than visible light and therefore higher resolving power. SEM emphasizes surfaces; TEM emphasizes intracellular ultrastructure.

Objective-driven tissue ID

Epithelia: classify by layers, then cell shape

LO 3-5

Epithelium	Morphology	Common recognition clue / role
Simple squamous	One layer of flattened cells with flattened nuclei	Thin exchange/lining surface. Endothelium lines vessels; mesothelium covers serosal surfaces.
Simple cuboidal	One layer; cells about as tall as wide; round central nuclei	Common in small ducts/tubules. Basement membrane is at the basal interface.
Simple columnar	One layer of tall cells; nuclei usually aligned toward the basal half	Absorption/secretion; goblet cells may be present.
Pseudostratified columnar	Nuclei at different levels, but all epithelial cells contact basement membrane	Often ciliated with goblet cells in respiratory tract.
Stratified squamous	Multiple layers; surface-most cells are flattened	Protection from abrasion; may be keratinized or nonkeratinized.

Basement membrane: extracellular matrix interface that anchors/supports epithelium, helps organize polarity, and forms a barrier between epithelium and underlying connective tissue. It may appear as a thin eosinophilic line in some H&E sections, but ultrastructure is best resolved by electron microscopy; PAS can accentuate basement membranes.

Invasion is a basement-membrane problem: epithelial dysplasia confined above the basement membrane is non-invasive; malignant epithelial cells that breach it are invasive carcinoma.

Type	Composition / appearance	Function / common setting
Loose (areolar)	Loose collagen/elastic fibers, abundant ground substance, many cells	Flexible support beneath epithelia; permits diffusion and immune-cell traffic
Dense irregular	Thick collagen bundles arranged in multiple directions	Resists stress from many directions; dermis/ capsules
Dense regular	Parallel collagen bundles with aligned fibroblasts	Strong unidirectional tensile resistance; tendons/ligaments
Elastic	Abundant elastic fibers / lamellae	Stretch and recoil; large elastic arteries and selected ligaments
Adipose	Large clear-appearing adipocytes; nucleus pushed to periphery in white fat	Energy storage, cushioning, insulation, endocrine signaling

Deck pp. 27-33

Histopathology language: recognize what changed

LO 7, 9

Adaptation

Hypertrophy = increased cell size. **Atrophy** = decreased cell size. **Hyperplasia** = increased cell number. **Metaplasia** = replacement of one mature cell type by another.

Neoplasia

Neoplasia is proliferation beyond normal regulation. A neoplasm is a tumor; behavior can be benign, premalignant/dysplastic, or malignant depending on the lesion.

Term	Meaning / morphology
Dysplasia	Disordered/abnormal growth with altered architecture, cytologic atypia, and increased turnover. Severe epithelial dysplasia may be termed carcinoma in situ.
Pleomorphism	Variation in cell/nuclear size and shape.
Anaplasia	Marked loss of differentiation; cells look very unlike their tissue of origin.
Hyperchromasia	Nucleus appears abnormally dark due to increased/coarse chromatin.
High N:C ratio	Large nucleus with relatively scant cytoplasm; common cytologic warning sign.
Invasion	Growth beyond normal confines, especially epithelial breach of basement membrane.
Desmoplasia	Reactive fibrotic stromal response that may accompany invasive carcinoma.
Metastasis	Spread of malignant cells to noncontiguous organs/tissues.

CARCINOMA LANGUAGE

Carcinoma = malignant epithelial neoplasm. **Adenocarcinoma** = malignant epithelial tumor showing gland formation. **Squamous cell carcinoma** = malignant tumor derived from squamous epithelium. **Adenoma** = gland-forming neoplasm, often benign or premalignant depending on context.

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Proscia: digital-slide workflow

LO 2

What you need to be able to do

- Open the assigned group/repository and select a digital slide.
- Use the thumbnail/minimap for rapid navigation.
- Zoom in/out with the magnification controls, mouse wheel, or trackpad gesture.
- Toggle annotations on/off and read the associated text.
- Move from low-power architecture to higher-power cellular detail, just as you would with a physical microscope.

Digital histology is still histology: begin with overall architecture. If you start at maximum zoom, you can lose the relationship between epithelium, basement membrane, stroma, and lesion boundaries.

Cervix / colposcopy / LEEP

Colposcopy examines the cervix for vascular and surface abnormalities. Acetic acid can produce acetowhite change in abnormal squamous epithelium. LEEP uses an electrified wire loop to excise a pathologic region for diagnosis/treatment.

LAST terminology

LSIL = low-grade squamous intraepithelial lesion with atypia limited to the lower portion of the epithelium. **HSIL** = high-grade lesion with atypia extending through a larger fraction of epithelial thickness.

Colon

Colonic adenomas are dysplastic/premalignant glandular lesions that often appear as polyps. Colonoscopy can both detect and remove them before invasion develops.

Direct tissue sampling

Biopsy, excision, LEEP, endoscopic biopsy/polypectomy, cytology, and surgical resection are examples of methods that provide tissue/cells for pathologic evaluation.

Trap 1: simple vs stratified is based on the number of layers contacting the surface, not how many nuclei you can see.

Trap 2: pseudostratified epithelium looks multilayered, but every epithelial cell reaches the basement membrane.

Trap 3: endothelium and mesothelium are specialized names for simple squamous epithelium based on location.

Trap 4: hematoxylin stains nucleic-acid-rich structures blue-purple; eosin stains protein-rich structures pink-red.

Trap 5: dysplasia/carcinoma in situ is not the same as invasion. Breach of the basement membrane changes the category.

Trap 6: hypertrophy = bigger cells; hyperplasia = more cells.

Trap 7: high magnification is not the first step. Architecture at low power prevents misidentification.

30-SECOND FINISH

Start low power. Think in 3D. Formalin → processing → paraffin → microtome → stain. H&E: nuclei blue-purple, proteins/collagen pink. PAS marks glycogen/mucin/basement membranes/fungi. Endothelium/mesothelium = simple squamous. Pseudostratified cells all touch basement membrane. Dense regular = parallel; dense irregular = multidirectional. Dysplasia is atypical growth; invasion crosses the basement membrane; metastasis spreads.